Lossless and lossy compression

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Applying the MDL principle in Machine learning

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For a smaller degree,
the code for f is short,
but the function does not predict
the values of Y well enough.
For larger degrees,
the data are perfectly predicted,
but the function f is too complex.
The degree 4 corresponds to the best
trade-off between the complexity of the model
and the quality of the prediction.

Algorithmic Information Theory is primarily about **lossless compression**. The point is to summarize objects in the most concise way, while keeping all the information needed to recover them. However, anomaly detection points to the possibility of getting rid of abnormal points, considered as mere noise. More generally, **lossy compression** consists of discarding noisy or contingent data in the hope of capturing the gist of a phenomenon. An extreme form of lossy prototype-based compression would be to keep the prototypes and to forget all data. This amounts to keeping only the model part in the expression of MDL. Simplification and sketching (as in comic strips) may be modelled as resulting from lossy compression.
Note that such a simplification can only occur after the MDL optimization has been performed.

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